MIL-C-81268A(OS)
18 February 1988
SUPERSEDING
Notice 2
29 May 1979

MILITARY SPECIFICATION

CLOTH, NYLON

This specification has been approved by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of polyvinyl chloride resin coated, 210 denier nylon cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current of the date of the solicitation

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Ordnance Station, Standardization/Documentation Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 8305

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PUBLICATIONS

American Society for Testing and Materials

ASTM D 1682	Standard Test Method for Breaking Load and Elongation of Textile Fabrics
ASTM D 1777	Method for Measuring Thickness of Textile Material
ASTM D 3776	Test Method for Mass Per Unit Area (Weight of Woven Fabrics)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3)

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in section 2 unless specified in the contract or order (see 6.2)

3.3 Physical properties. The physical properties shall be as specified in Table I

TABLE I Physical properties

Requirement	Average values Minimum	Maximum
Thread count, threads per inch		
Warp	22	24
Fill	22	24
Breaking strength, pounds		
Warp	65	—
Fill	65	—
Elongation, percent		
Warp	—	35
Fill	—	35
Thickness, inches	0.006	0.009
Weight, ounces per square yard	1.55	2.00
Weave	plain	

3.4 Workmanship. The material shall be uniform in weave and composition and shall be free of visible damage, foreign material, or other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of contract Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material

4.2 Classification of inspections Inspection of the nylon cloth shall be classified as follows

a First article inspection (see 4 4)
b Quality conformance inspection (see 4.5)

4.3 Sampling.

4.3.1 First article sample. A first article sample of three test specimens of the nylon cloth, manufactured in accordance with 3.1 shall be subjected to the first article tests detailed in 4 4 at an activity designated by the procuring activity Further production of the nylon cloth by the supplier, prior to the approval of the first article sample, shall be at the supplier's risk

4.3.2 Quality conformance inspection sampling. Except as otherwise specified, the sampling plans and procedures set forth in MIL-STD-105, level II shall be used in the determination of acceptability of the nylon cloth inspected as specified in 4.5 From each end of the roll of material for sampling, select ten specimens, 4 inches in width and 6 inches in length Two sets of five specimens from each end shall be taken in the following manner one set of five for warp breaking strength shall have the longer dimension parallel to the warp yarns; the other set for fill breaking strength shall have the longer dimension parallel to the fill yarns No two specimens within a set of five for warp breaking strength shall contain the same warp yarns or for fill breaking strength shall contain the same fill yarns Specimens shall be taken no nearer to the selvage than one-tenth the width of the fabric, nor nearer than one foot from the ends of the rolls.

4.3.3 Inspection lot. An inspection lot of nylon cloth shall be all the material presented for acceptance at one time and produced in a single manufacturing run under homogeneous conditions of manufacture

4.4 First article inspection. The first article sample shall satisfactorily pass the quality conformance inspections detailed in 4.5

4.5 Quality conformance inspection.

4.5.1 Visual examination. Visually examine all containers or rolls in a lot to determine conformance to 3.4 and Section 5

4.5.2 Physical tests.

4.5.2.1 Thread count test The total number of strands for 3 inches in both the warp and fill directions shall be visually determined for each specimen.

4.5.2.2 Breaking strength and elongation tests. All tensile property tests shall be performed using the modified grab test method of ASTM D 1682, and a constant rate of extension of one inch per minute All test samples shall be conditioned in a desiccator containing anhydrous magnesium perchlorate or anhydrous calcium sulfate at 75° ± 5°F for a minimum of 24 hours immediately prior to testing A stress-strain curve shall be recorded for each tensile test Elongation shall be expressed as the percentage increase in length of the sample as determined from the stress-strain curve Breaking strength shall be read directly from the test apparatus

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4.5.2.3 Thickness test. The thickness of the fabric shall be determined with a suitable testing instrument having a presser foot 0.375 inch in diameter at 1 1 ± 0 1 pounds per square inch (psi) for 5 seconds, and in accordance with ASTM D 1777

4.5.2.4 Weight test. The fabric weight shall be determined in accordance with ASTM D 3776 The specimens shall be conditioned and the weight in ounces per square yard calculated as specified therein

4.6 Acceptance criteria.

4.6.1 First article. Failure of any sample to comply with any requirement of this specification shall be cause for rejection of the sample

4.6.2 Quality conformance. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the sample Acceptance of the lot shall be based upon an Acceptable Quality Level (AQL) of 1 0 percent in accordance with MIL-STD-105, level II, normal inspection, single sampling

5. PACKAGING

5.1 Packaging and marking. Preservation, packaging and packing shall be in accordance with MIL-P-116, method III

5.2 Marking. Marking shall be in accordance with MIL-STD-129 Marking shall include, but not be limited to, the following information

a Manufacturer's name and location
b Material trade name
c Lot number, batch number and date of manufacture
d Number and revision letter of this specification

6. NOTES

6.1 Intended use. Material purchased under this specification is intended for use as an inhibitor or the external surface of propellant grain in rocket motors

6 2 Acquisition requirements. Acquisition documents should specify, but not be limited to, the following information.

a Title, number and revision letter of this specification
b Special preservation, packaging and packing requirements, if applicable
c Length and width of rolls
d Place of inspection
e Place of delivery
f Whether first article sample is required
g Request for test data

6.3 First article. When a first article inspection is required, the item should be a first article sample The first article should consist of three test specimens of the nylon cloth The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations approval of first article test results and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract

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6.4 Subject term (key word) listing

Cloth, Nylon
Propellant wrapper
Propellant inhibitor
Rocket motor

Custodian.
Navy OS

Preparing activity
Navy OS

(Project No 8305N171)

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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